Table 1. Cell Sources for Salivary Gland Organoids

Cell source	Accessibility & expansion	Key advantages	Key limitations	Applications	Refs
Adult epithelial cells	Readily accessible; salivary-lineage biased; limited expansion affected by age or pathology	Autologous use; native gland identity	No single defined stem-cell hierarchy; variable establishment and differentiation	Disease modeling; regeneration; drug screening	13–15, 23, 30–32, 79, 88
Pluripotent stem cells (iPSCs/ESCs)	iPSCs from somatic cells; ESCs from embryos; extensive expansion; directed differentiation required	Genetic and rare-disease modeling; potential access to major-gland lineages	Incomplete mature function; lengthy and costly differentiation; protocol-dependent engineering	Developmental and genetic disease modeling; viral tropism	33–40, 47
Embryonic and fetal tissues	Limited availability; abundant native progenitors; high proliferation and differentiation potential	Robust branching morphogenesis; benchmarks for PSC differentiation	Ethical and supply constraints; incomplete ductal architecture	Developmental biology; reference standards	41–46

An integrated approach matching each cell source to the best-suited research question provides the most comprehensive platform.

Abbreviations: iPSC, induced pluripotent stem cell; ESC, embryonic stem cell; PSC, pluripotent stem cell.